

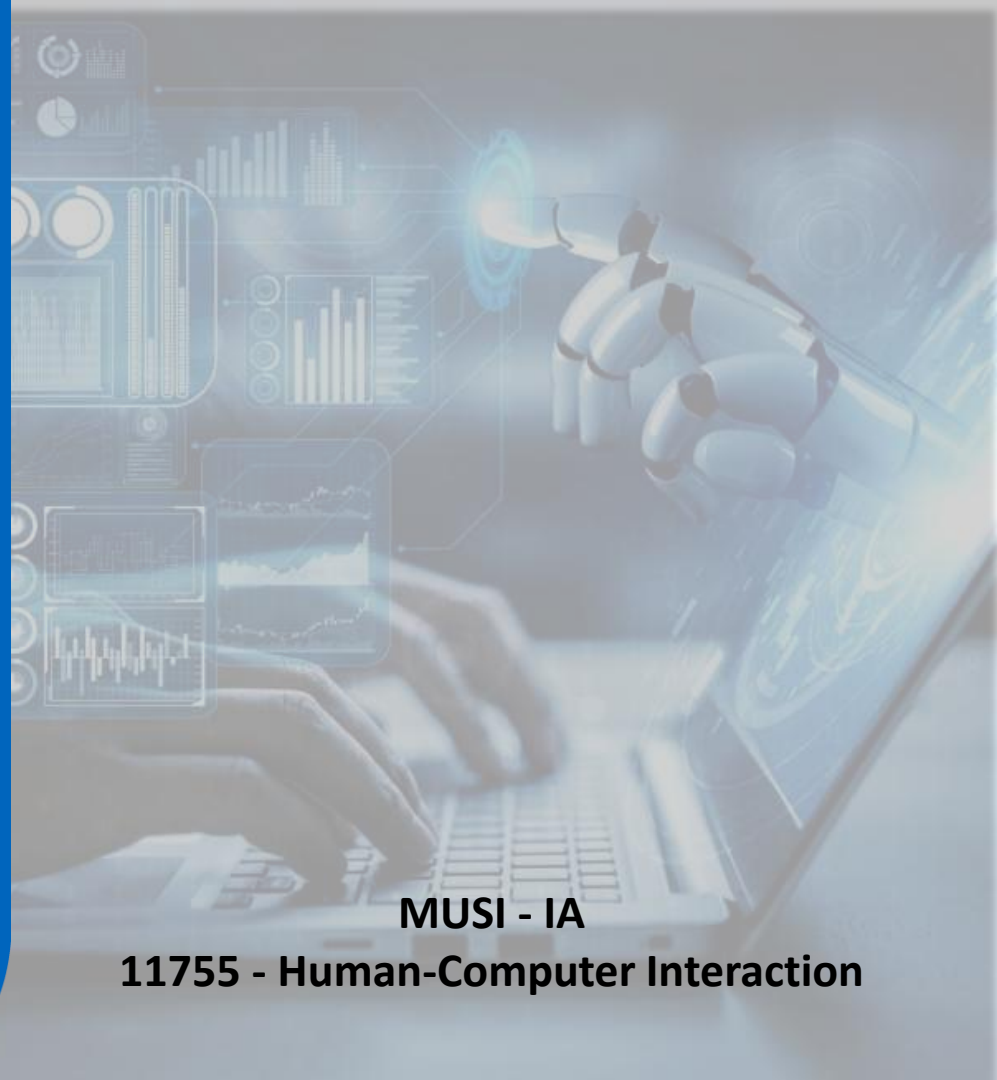


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01 - Introduction to Human-Computer Interaction (HCI)

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3-3-2026



MUSI - IA

11755 - Human-Computer Interaction



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1

What is HCI?

Alan Dix

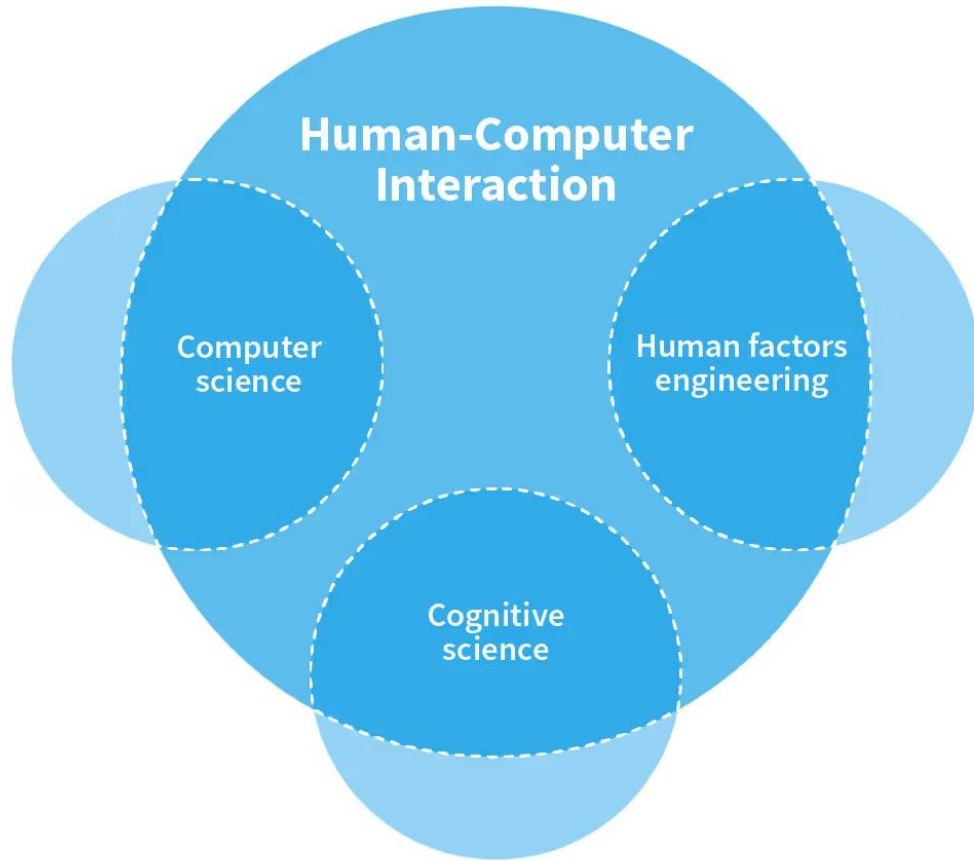
Professor, Expert in
Human-Computer
Interaction



What is HCI?

- One side of HCI is the **academic discipline** and the other is the **applied design discipline**, which is about how you create interventions with technology that make a difference to people.
- What would I hope you would learn from studying HCI?
 - ... bare facts. Facts about the nature of computers. There's facts about the nature of human psychology, physiology – a whole range of things—social interactions.
 - ... is about analysis. It's about looking at a situation and trying to make sense of it: what's going on there.
 - ... how you make sure that this wonderful system that you designed actually gets used by people in the real world and makes a real difference to people.

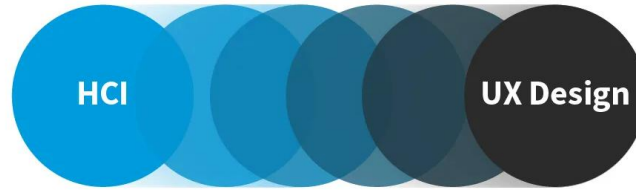
The Multidisciplinary Field of HCI



Human-computer interaction (HCI) is a multidisciplinary field of study focusing on the design of computer technology and, in particular, the interaction between humans (the users) and computers. While initially concerned with computers, HCI has since expanded to cover almost all forms of information technology design.

Alan Dix. “What is Human-Computer Interaction (HCI)?” [Interaction Design Foundation - IxDF](https://www.interaction-design.org/fundamentals/what-is-human-computer-interaction-hci).

The Evolution of UX Design



Interaction Design Foundation
interaction-design.org

- HCI is a broad field which overlaps with areas such as **user-centered design (UCD)**, **user interface (UI) design** and **user experience (UX) design**. In many ways, **HCI** was the forerunner to **UX design**.
- Practitioners of HCI tend to be more academically focused. They're involved in scientific research and developing empirical understandings of users.
- Conversely, UX designers are almost invariably industry-focused and involved in building products or services—e.g., smartphone apps and websites.

“What is Human-Computer Interaction (HCI)?” [Interaction Design Foundation - IxDF](http://interaction-design.org).

What is HCI?

Theoretical and disciplinary influences:

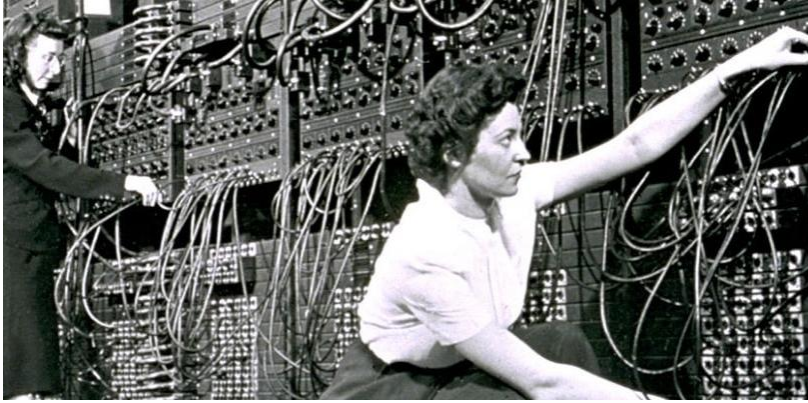
- HCI draws from multiple disciplines including computer science, psychology, ergonomics and social sciences.
- In the early days, the main disciplines involved were computer science, psychology, and ergonomics.
- Later, social sciences, particularly the ethnographic and anthropological aspects of sociology, became important.
- Ergonomics initially focused on physical ergonomics, but this declined as computers became more commoditized. However, physical design issues have resurfaced with the rise of tangible computing and product design.



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The history and evolution of HCI

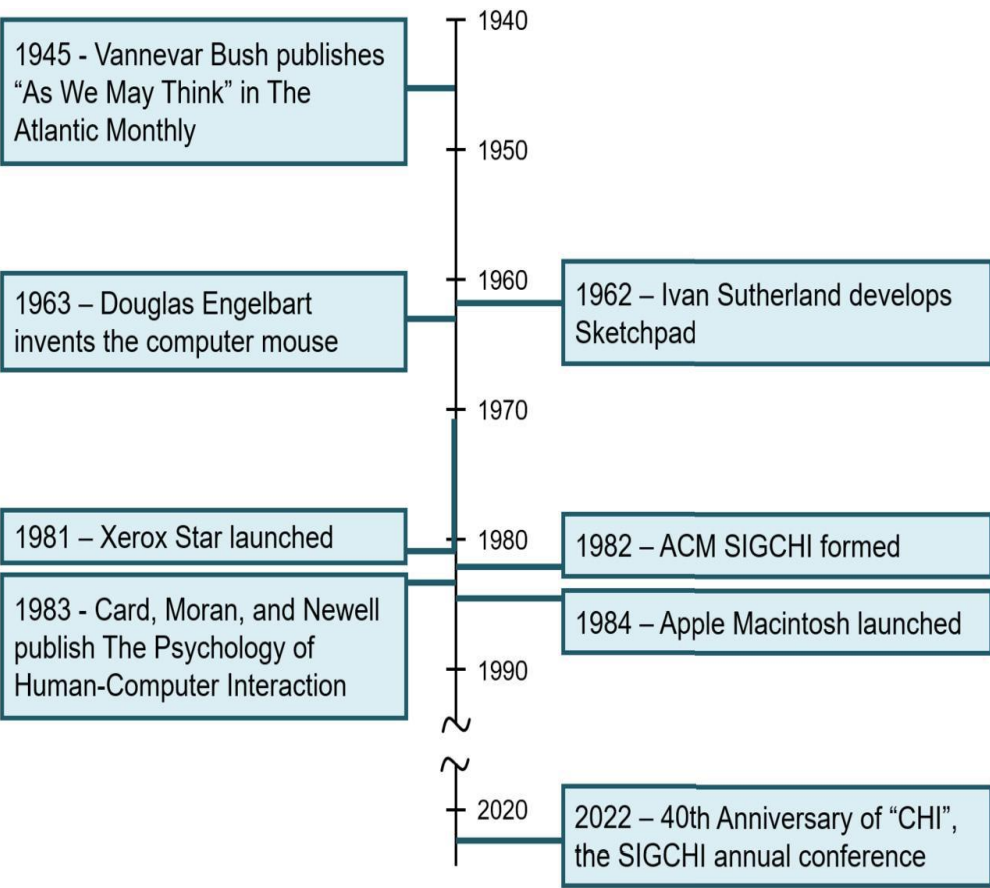


ENIAC (Electronic Numerical Integrator and Computer) was the first [programmable](#), electronic, general-purpose digital computer, completed in 1945.



The [Apple Vision Pro](#) is a mixed-reality headset developed by Apple. It became available for purchase on February 2, 2024, in the United States.

79 years



“In the beginning, there were humans. In the 1940s came computers. In the 1980s came interaction. Wait! What about 1940 to 1980?

Interaction was not on the minds of the engineers and scientists who designed, built, configured, and programmed early computers.

But, by the 1980s, interaction arrived. The new computers were not only powerful, **they were usable – by anyone!**”

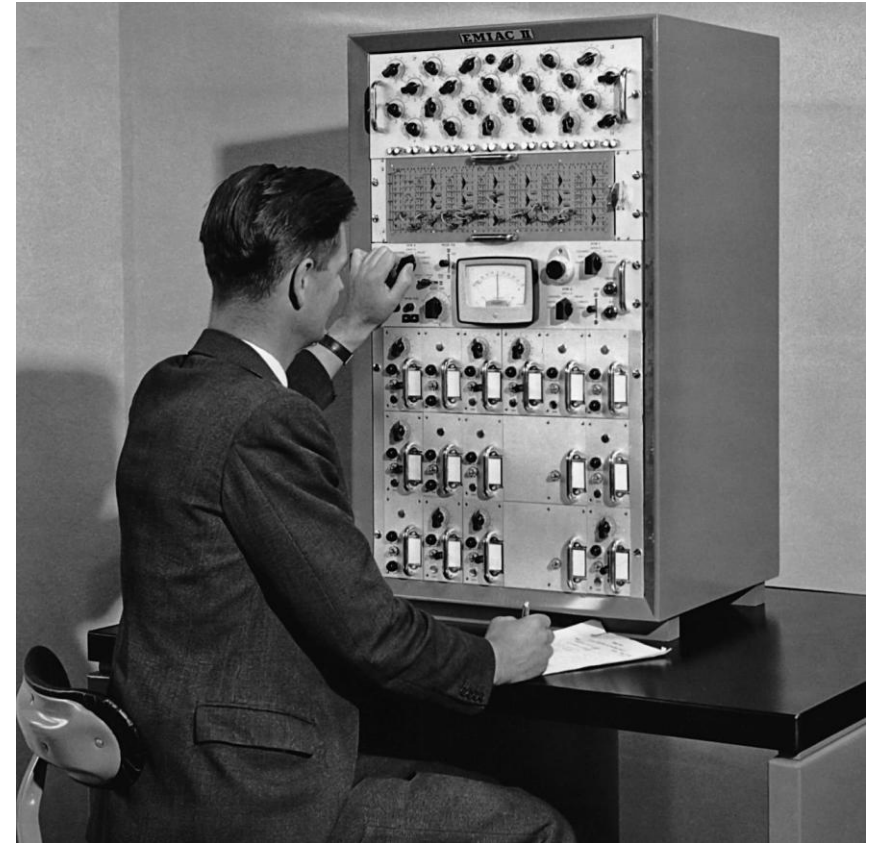
The history and evolution of HCI

- **Direct Manipulation:** The idea of directly manipulating visible objects on the screen using a pointing device was first shown in [Ivan Sutherland's](#) Sketchpad. The mouse was developed at Stanford Research Laboratory by [Douglas Engelbart](#) in 1963.
- **Windows:** Multiple tiled windows were demonstrated in Engelbart's [NLS](#) in 1968. Overlapping windows were proposed by [Alan Kay](#) and first appeared in his Smalltalk system in 1974.
- **Hypertext:** The idea of linked documents is credited to [Vannevar Bush](#), with [Ted Nelson](#) coining the term "hypertext" in 1965. Engelbart's NLS system also made extensive use of linking.

The history and evolution of HCI

“However, the very first true HCI paper dates back into the late 1950s, with Shackel’s [‘Ergonomics for a Computer’](#) [112] . While Sutherland and Engelbart were early examples of the vision/innovation side of HCI, Shackel’s first HCI paper came more from a practical design perspective, the redesign of the control panel of EMIAC II, an early analogue computer.”

A. Dix, “Human–computer interaction, foundations and new paradigms,” *Journal of Visual Languages & Computing*, vol. 42, pp. 122–134, Oct. 2017, doi: 10.1016/j.jvlc.2016.04.001.



And Then Steve Said, ‘Let There Be an iPhone’



Share full article



314

“It’s hard to overstate the gamble Jobs took when he decided to unveil the iPhone back in January 2007. Not only was he introducing a new kind of phone — something Apple had never made before — he was doing so with a prototype that barely worked.”

Fred Vogelstein, [The New York Times Magazine](#). October 2013



The history and evolution of HCI

Weekly Assignment 1: There is a time-gap in Fig. 1.1 (previous slide) from 1984, when the Apple Macintosh was launched, to 2022, when SIGCHI celebrated its 40th anniversary. Pick five significant events (e.g., product releases) in this time frame that significantly impacted computing and, in particular, human-computer interaction. Ensure at least one event is in each decade in the 1990s, 2000s, and 2010s. Prepare a brief scientific article (3 pages maximal) highlighting the event and its importance to the field. Aim to juxtapose the significance of the event with computing practice around the time of the event. Perhaps include statistics, such as sales or other data, that confirm the significance of the event.

For the article use the template (Word or LaTeX) for IEEE TRANSACTIONS and JOURNALS provided in the course site (also downloadable [here](#) by choosing Transactions, Journals and Letters in the first step and IEEE Communications Surveys and Tutorials in the second step)



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3

Usability and user experience (UX)

Rethinking a TV Remote for the Elderly



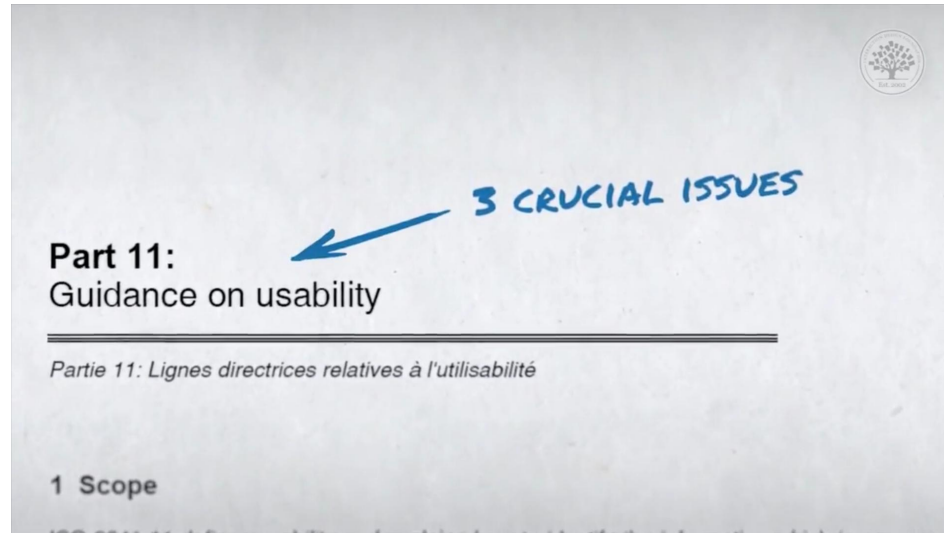
“Evaluation of the gathered interview data:

- Too many tiny buttons on the remote confuse them and they have a tough time pressing buttons with precision.
- Majority of the respondents hold their remotes in one hand and operate it with the other.
- I learned that the elderly are creatures of habit and have a fixed set of favorite channels. On an average 5 to 6 channels, that is the only TV they watch.”

Definition of Usability

Usability is the extent to which a product can be used by specified users to achieve specified goals with three things: effectiveness, efficiency and satisfaction, in a specified context of use.

Interaction Design Foundation - IxDF. [“What is Usability?”](#) and [ISO 9241-210:2019 Ergonomics of human-system interaction, Part 210: Human-centred design for interactive systems](#), 3.13



3 CRUCIAL ISSUES

Part 11: Guidance on usability

Partie 11: Lignes directrices relatives à l'utilisabilité

1 Scope

ISO 9241-11 defines usability and explains how to identify the information which is necessary

Definition of Usability (quality components)

Usability is defined by 5 quality components:

- **Learnability:** How easy is it for users to accomplish basic tasks the first time they encounter the design?
- **Efficiency:** Once users have learned the design, how quickly can they perform tasks?
- **Memorability:** When users return to the design after a period of not using it, how easily can they reestablish proficiency?
- **Errors:** How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
- **Satisfaction:** How pleasant is it to use the design?

Definition of Usability (other attributes)

There are many other important quality attributes. A key one is utility, which refers to the design's functionality: Does it do what users need?

Usability and utility are equally important and together determine whether something is useful: It matters little that something is easy if it's not what you want. It's also no good if the system can hypothetically do what you want, but you can't make it happen because the user interface is too difficult.

- Definition of **Utility** = whether it provides the **features you need**.
- Definition of **Usability** = how **easy & pleasant** these features are to use.
- Definition of **Useful** = usability + utility.

Definition of Usability (another view of quality components)

Your design's usability should contain these elements:

- **Effectiveness**—It supports users in completing actions accurately.
- **Efficiency**—Users can perform tasks quickly through the easiest process.
- **Engagement**—Users find it pleasant to use and appropriate for its industry/topic.
- **Error Tolerance**—It supports a range of user actions and only shows an error in genuine erroneous situations.
- **Ease of Learning**—New users can accomplish goals easily and even more easily on future visits.

Definition of Usability



Nielsen Norman Group

- **Learnability**
- **Efficiency**
- **Memorability**
- **Errors**
- **Satisfaction**

Jacob Nielsen, [Usability 101: Introduction to Usability](#)



INTERACTION DESIGN
FOUNDATION

- **Effectiveness**
- **Efficiency**
- **Engagement**
- **Error Tolerance**
- **Ease of Learning**

Interaction Design Foundation - IxDF. [“What is Usability?”](#)

Definition of Usability

Is usability the same as usefulness?

No, usability and usefulness are distinct concepts in design. Usability refers to how easily and effectively users can interact with a product, and how satisfied they are with that interaction. On the other hand, usefulness addresses whether the product fulfills a specific need or solves a meaningful set of problems for users. A product can be usable (easy to use) but not useful (doesn't meet user needs).

Interaction Design Foundation - IxDF. [“What is Usability?”](#)

Recommended Reading: [Useful, Usable, and Used: Why They Matter to Designers](#)

Definition of User Experience (UX)

The first requirement for an exemplary user experience is to meet the exact needs of the customer, without fuss or bother. Next comes simplicity and elegance that produce products that are a joy to own, a joy to use. True user experience goes far beyond giving customers what they say they want, or providing checklist features. In order to achieve high-quality user experience in a company's offerings there must be a seamless merging of the services of multiple disciplines, including engineering, marketing, graphical and industrial design, and interface design.

D. Norman and J. Nielsen, [The Definition of User Experience \(UX\)](#)

Definition of User Experience (UX)

The term “UX” was invented by Donald Norman, a cognitive psychologist who started working at Apple in the early 1990s. He took on the role of User Experience Architect, making him the first (known) person in history to have UX in their job title.

Explaining the origins of UX, Donald Norman said:

“I invented the term because I thought human interface and usability were too narrow. I wanted to cover all aspects of the person’s experience with a system, including industrial design, graphics, the interface, the physical interaction and the manual.”

C Vinney, [What is UX design?](#)

NN/g



Don Norman: The Term UX

What is User Experience (UX) Design?

User experience (UX) design is the process design teams use to create products that provide meaningful and relevant experiences to users. UX design involves the design of the entire process of acquiring and integrating the product, including aspects of branding, design, usability and function.

“No product is an island. A product is more than the product. It is a cohesive, integrated set of experiences. Think through all of the stages of a product or service – from initial intentions through final reflections, from the first usage to help, service, and maintenance. Make them all work together seamlessly.” Don Norman

What is User Experience (UX) Design?

UX designers can—and do—use a process called "**user-centered design**." UX design is the discipline behind what—or rather, who—ensures that a product is easy to use and enjoyable for the people who access, use, and are (hopefully) delighted at what brands produce for them.

Accessibility, information hierarchy, responsive design, and navigation flows are part of UX—and they're big considerations that call for frequent collaboration with the **UI designers** so they can make sure that the product works for the most users possible.

Interaction Design Foundation - IxDF. [“What is User Experience \(UX\) Design?”](#)

User Experience Design



What is User Experience (UX) Design?

As a UX designer, you should consider the **Who, Why, What and How** of product use. The **Why involves the users' motivations for adopting a product**, whether they relate to a task they wish to perform with it or to values and views that users associate with the ownership and use of the product. The **What addresses the things people can do with a product**—its functionality. Finally, the **How relates to the design of functionality in an accessible and aesthetically pleasant way**.

Interaction Design Foundation - IxDF. [“What is User Experience \(UX\) Design?”](#)



**Garrett's
5 Elements
of User
Experience**

**Strategy
Scope
Structure
Skeleton
Surface**

UX vs UI: What's the Difference?

“User Experience Design” is often used interchangeably with terms such as “User Interface Design” and “Usability.” However, while usability and user interface (UI) design are important aspects of UX design, they are subsets.

A UX designer is concerned with the entire process of acquiring and integrating a product, including aspects of branding, design, usability and function. The story begins before the device is even in the user's hands.

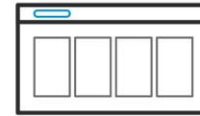
Interaction Design Foundation - IxDF. [“What is User Experience \(UX\) Design?”](#)

UX vs UI: What's the Difference?

UI designers build interfaces—namely, “**user interfaces**”—for digital products or services that are centered on detailed interaction. **UI is more specialized than UX design**, and that's because it focuses—as in, exclusively focuses—on the elements that users know, such as buttons and icons. **UI elements do tend to be visual**, to be sure; that said, though, **there are other types of UI, such as voice- and gesture-based interfaces.**

Interaction Design Foundation - IxDF. [“What is User Experience \(UX\) Design?”](https://www.interaction-design.org/)

Common Types of User Interface



Graphical User Interfaces (GUIs)



Voice-controlled Interfaces (VUIs)



Gesture-based Interfaces

Interaction Design Foundation
[interaction-design.org](https://www.interaction-design.org/)



UX | UI



UX vs UI: What's the Difference?



Key Difference Between UX and UI

UX Design	VS	UI Design
Focuses on the holistic experience of the user.		Focuses on the specific visual touchpoints of the user.
Centers on strategy, structure, and interaction design.		Centers on surface-level aspects of design, including visuals.
Involves studying the user's journey, and designing information architecture.		Involves designing the tangible elements of the experience, such as the choice of elements and how they interact.
Outputs include personas, user journey maps, wireframes, prototypes.		Outputs include mockups, high-fidelity layouts, and prototypes.
Incorporates a user's environment, mood, and context.		It is more tangible as users directly interact with the UI.

What is User Centered Design (UCD)?

User-centered design (UCD) is an iterative design process in which designers focus on the users and their needs in each phase of the design process. In UCD, design teams involve users throughout the design process via a variety of research and design techniques, to create highly usable and accessible products for them.

In user-centered design, designers use a mixture of investigative methods and tools (e.g., surveys and interviews) and generative ones (e.g., brainstorming) to develop an understanding of user needs. The term was coined in the 1970s. Later, cognitive science and user experience expert Don Norman adopted the term in his extensive work on improving what people experience in their use of items.

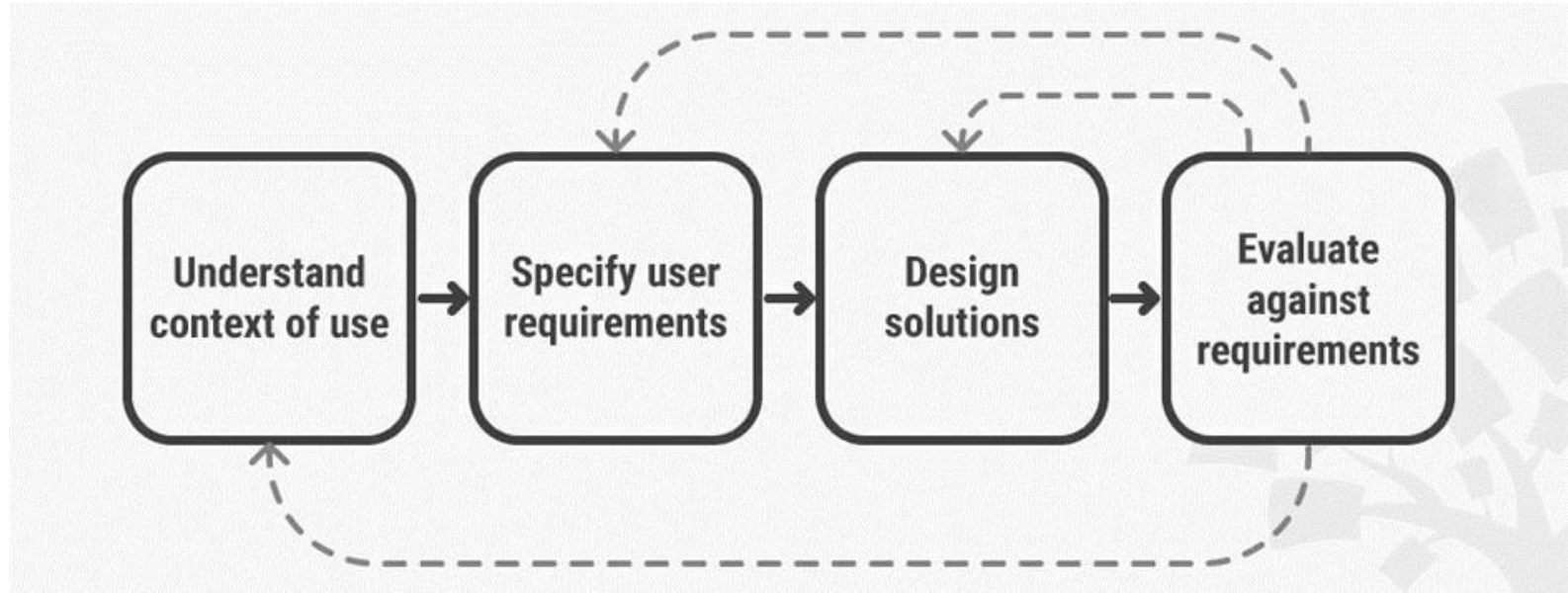
What is User Centered Design (UCD)?

4 Phases in UCD

Generally, each iteration of the UCD approach involves **four distinct phases**. First, as designers working in teams, we try to **understand the context in which users may use a system**. Then, we **identify and specify the users' requirements**. A **design phase** follows, in which the design team develops solutions. The team then proceeds to an **evaluation phase**. Here, you assess the outcomes of the evaluation against the users' context and requirements, to check how well a design is performing. From here, your team makes further iterations of these four phases, and you continue until the evaluation results are satisfactory.

What is User Centered Design (UCD)?

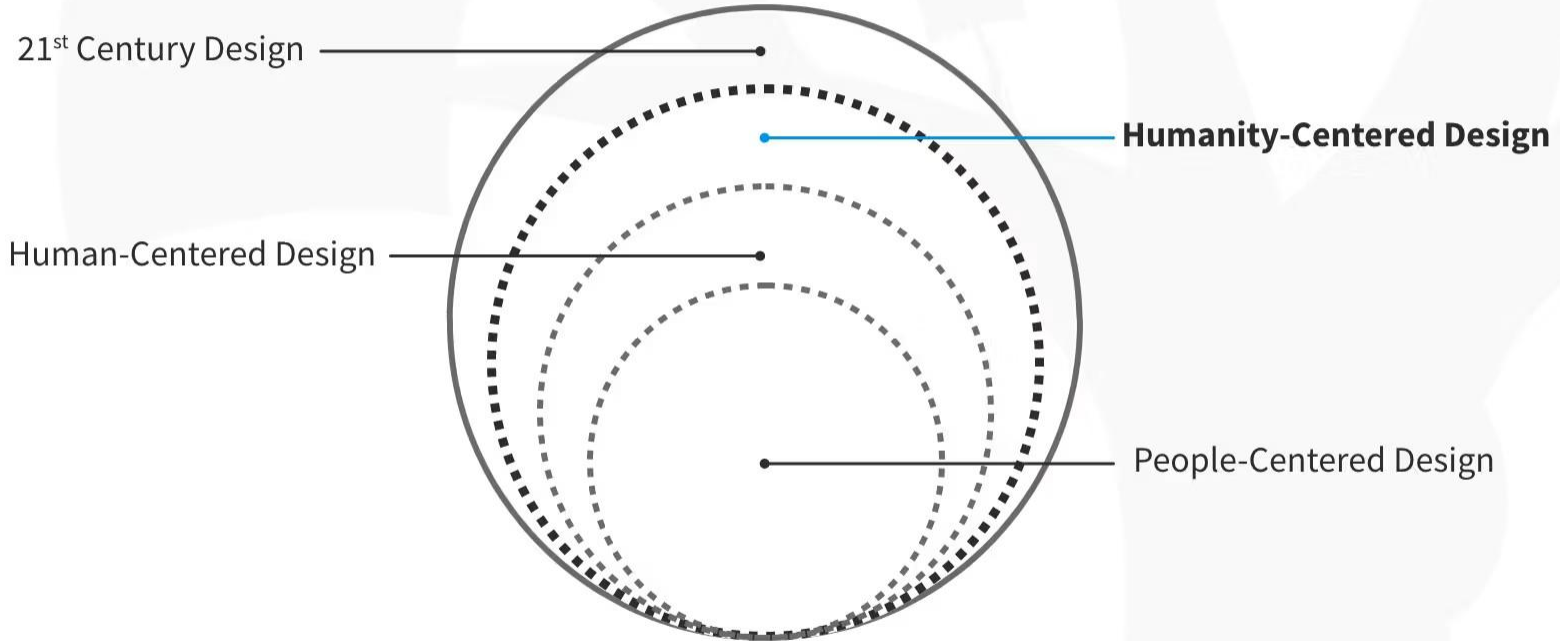
4 Phases in UCD



An aerial night photograph of a city, likely Los Angeles, showing a complex highway interchange with multiple overpasses and ramps. The city is densely packed with buildings, many of which are illuminated with lights, creating a vibrant, glowing effect against the dark night sky. The overall scene conveys a sense of urban complexity and infrastructure.

Design for the 21st Century with Don Norman

Humanity-Centered Design



Recommended Readings:

- [1] A. Dix, J. Finlay, G. D. Abowd, and R. Beale, Human-computer interaction, 3rd edition. Pearson Education Limited, 2004. (Introduction)
- [2] H. R. Hartson and P. S. Pyla, The UX book: Agile UX design for a quality user experience, Second edition. Morgan Kaufmann, 2019. (Chapter 1)
- [3] I. S. MacKenzie, Human-computer interaction: an empirical research perspective, Second edition. Morgan Kaufmann, 2024. (Chapter 1)
- [1] D. A. Norman, The design of everyday things, Rev. and Expanded edition. Basic Books, 2013. (Chapter 1)



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